Mean, Median, and Baselines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Mean, Median, and Baselines before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Mean, Median, and Baselines trace and record the first step you can explain without help.

Use W2: Score separately for Constant **prediction** baselines. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Mean, Median, and Baselines explanation against the supplied facts before discussing it.

Use the worked case for Constant **prediction** baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Mean, Median, and Baselines target to a trace and a justified conclusion.

For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Mean, Median, and Baselines case and show the deciding evidence.

Training heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

Compute both and **training absolute error**s.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

For **training** heights 2, 3 and 10 cm, calculate **mean** and **median**. Which constant is less influenced by the high value?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 20 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Mean, Median, and Baselines adds to Responsible Plant Classifier and keep the evidence.

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Fit two constants

Fit two constants. Apply the supplied Mean, Median, and Baselines facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Mean, Median, and Baselines", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Find the **mean** and **median** of the worked **training** targets.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Score separately

> Score separately. Apply the supplied Mean, Median, and Baselines facts and keep your reasoning.

Calculate MAE for both baselines on [4, 8].

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Inspect an extreme value

> Inspect an extreme value. Apply the supplied Mean, Median, and Baselines facts and keep your reasoning.

Replace 12 by 40. What changes?

My reasoning and evidence

W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Mean, Median, and Baselines facts and keep your reasoning.

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

My reasoning and evidence

W5 Prevent leakage

> Prevent leakage. Apply the supplied Mean, Median, and Baselines facts and keep your reasoning.

> Why not predict the evaluation **mean** as the baseline?

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Mean, Median, and Baselines

> Fit two constant height baselines and compare them on held-out examples. Explain the labelled step before checking the changed condition.

Fit two constant height baselines and compare them on held-out examples.

Training heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/20

Mean, Median, and Baselines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fit mean

Training mean=(2+4+6+20)/4=8 cm.

↓

2. Fit median

Training median=(4+6)/2=5 cm.

↓

3. Evaluate constants

Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm.

CHANGE ONE THING

Replace training 20 with 8 while keeping evaluation unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Fit two constant height baselines and compare them on held-out examples.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

1. Fit **mean**

2. Fit **median**

3. Evaluate constants

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Evaluate constants **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace **training** 20 with 8 while keeping evaluation unchanged. Which conclusion is supported by the supplied facts?

1. Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Repair a missing building block

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score **mean** every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

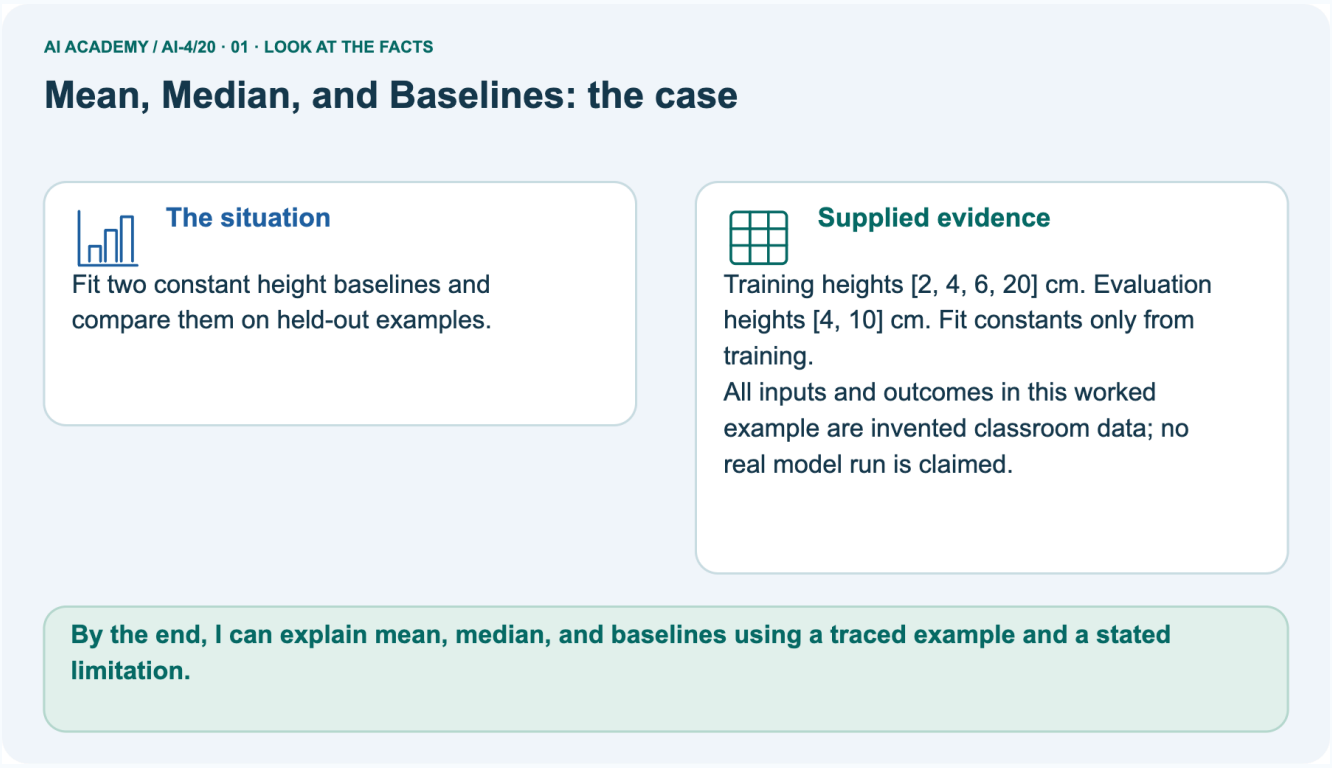
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Mean, Median, and Baselines picture from its labels, then record a relevant limit.

01. Mean, Median, and Baselines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

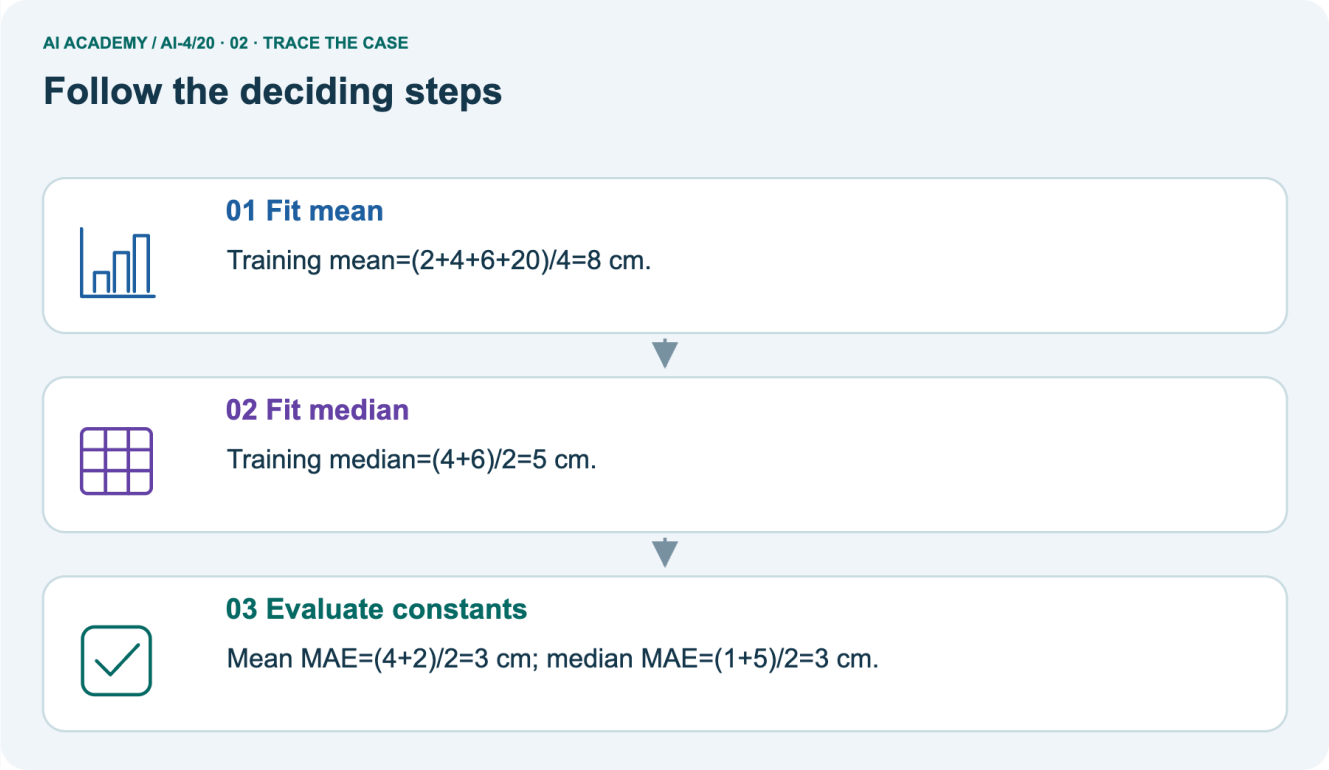
Fit two constant height baselines and compare them on held-out examples. Supplied facts: **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



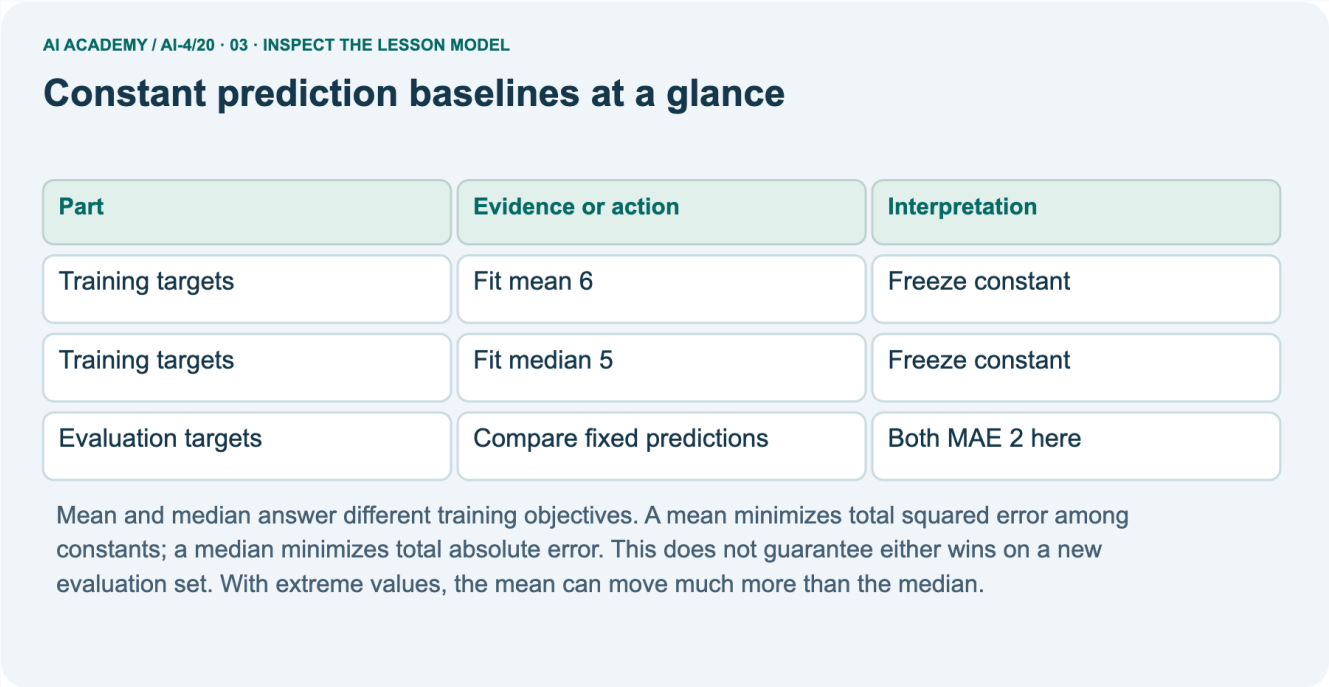
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **mean**: **Training mean**=(2+4+6+20)/4=8 cm. → Fit **median**: **Training median**=(4+6)/2=5 cm. → Evaluate constants: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Constant prediction baselines at a glance



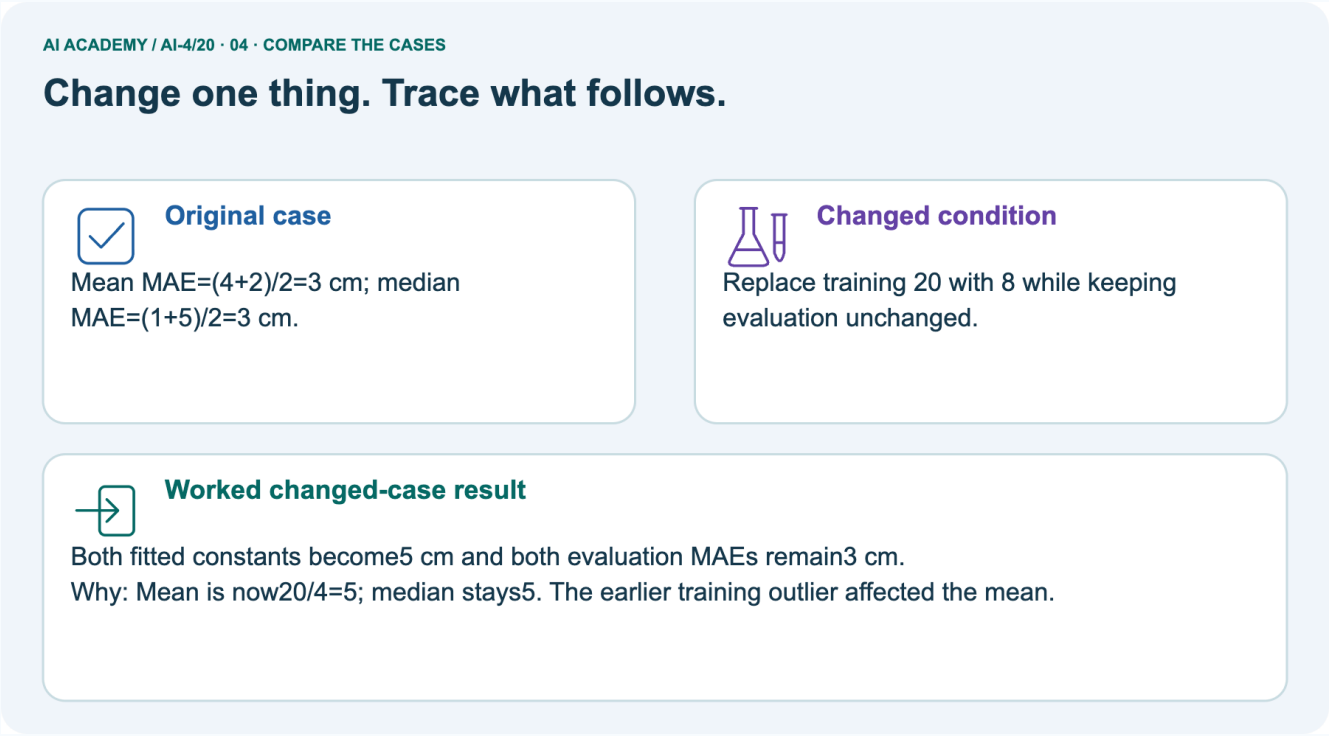
Follow the labelled picture with the matching explanation. Open the vector diagram

Constant **prediction** baselines at a glance. **Training** targets | Fit **mean** 6 | Freeze constant. **Training** targets | Fit **median** 5 | Freeze constant. Evaluation targets | Compare fixed **prediction**s | Both MAE 2 here

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

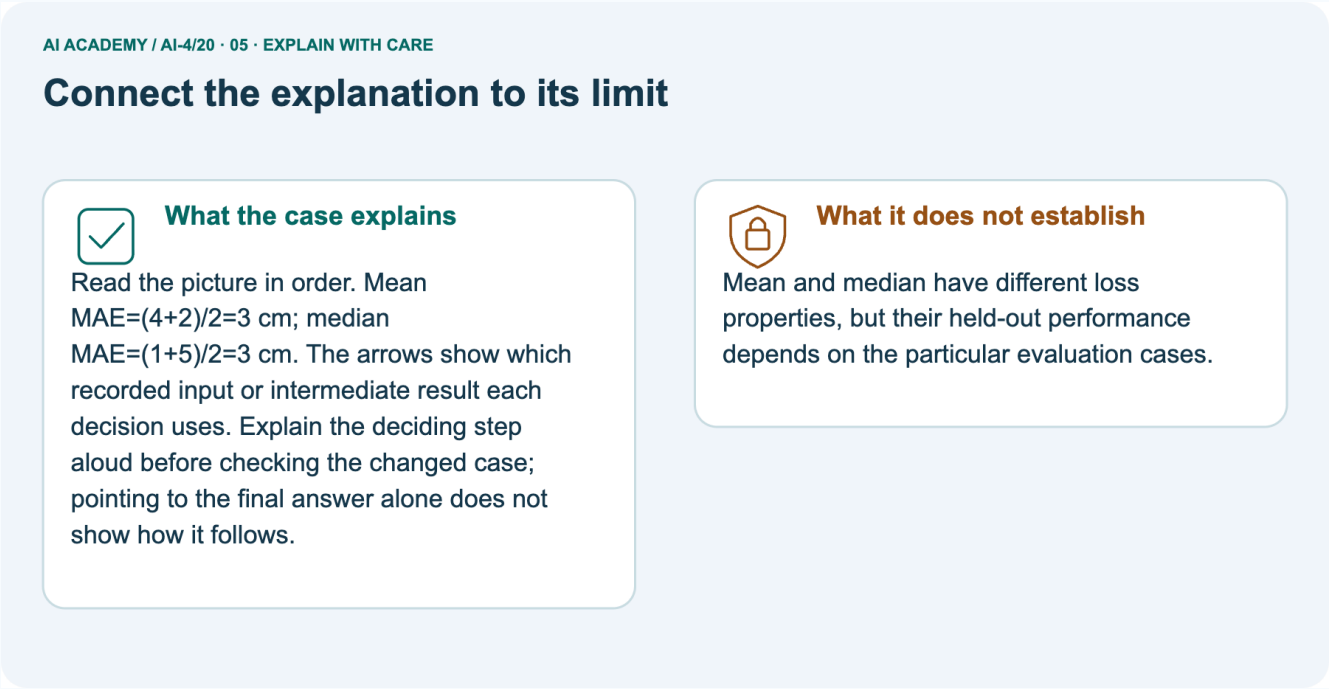
Original case: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Changed condition: Replace **training** 20 with 8 while keeping evaluation unchanged.

Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.
Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

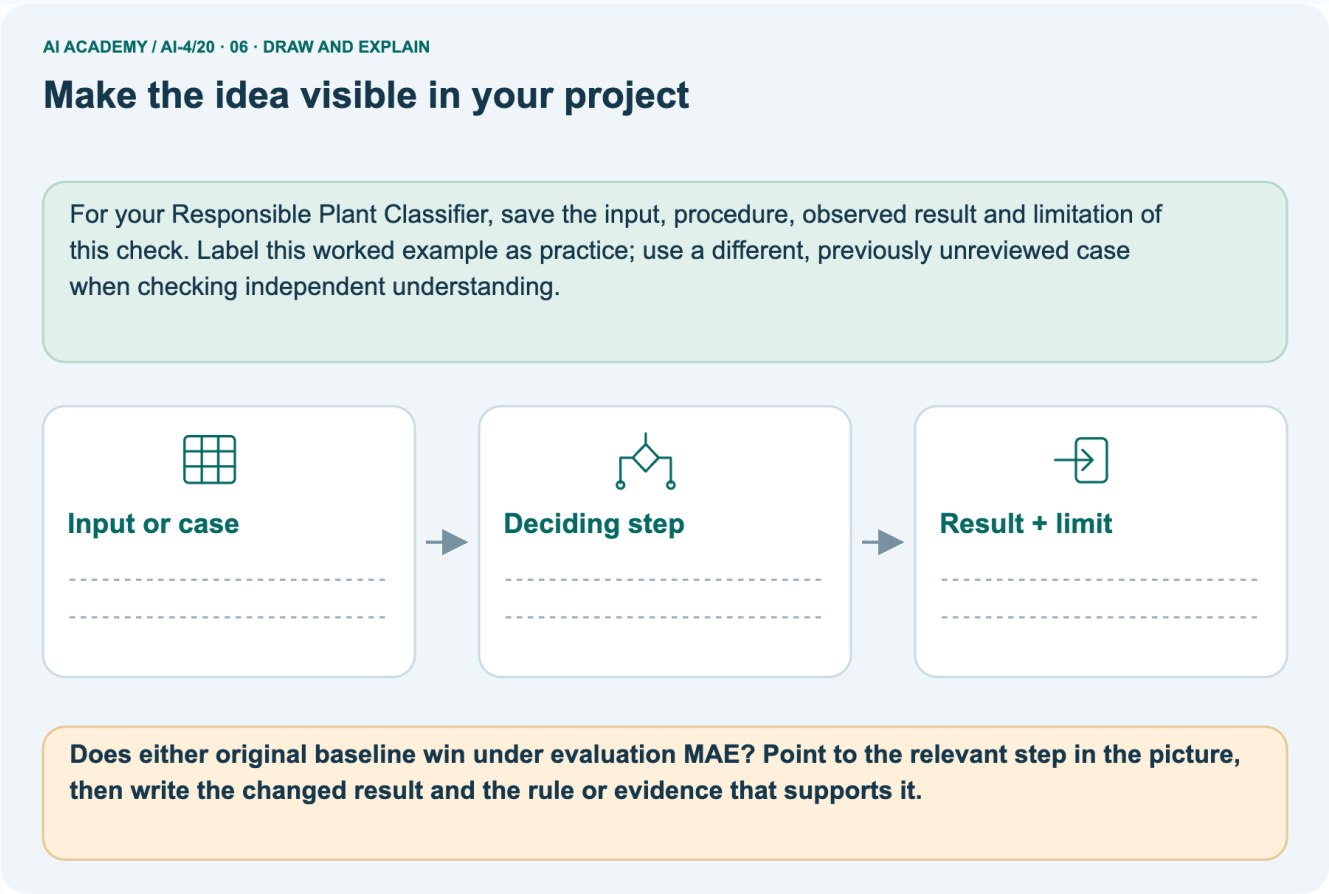
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.